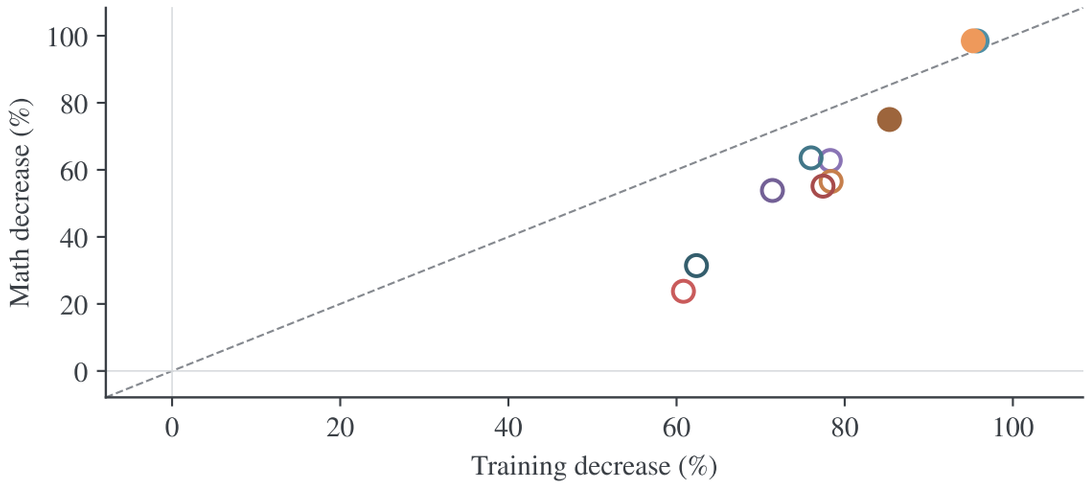
Llama-3.2-3B-Instruct | Training pool vs. held-out

Relative RKL decrease (%)

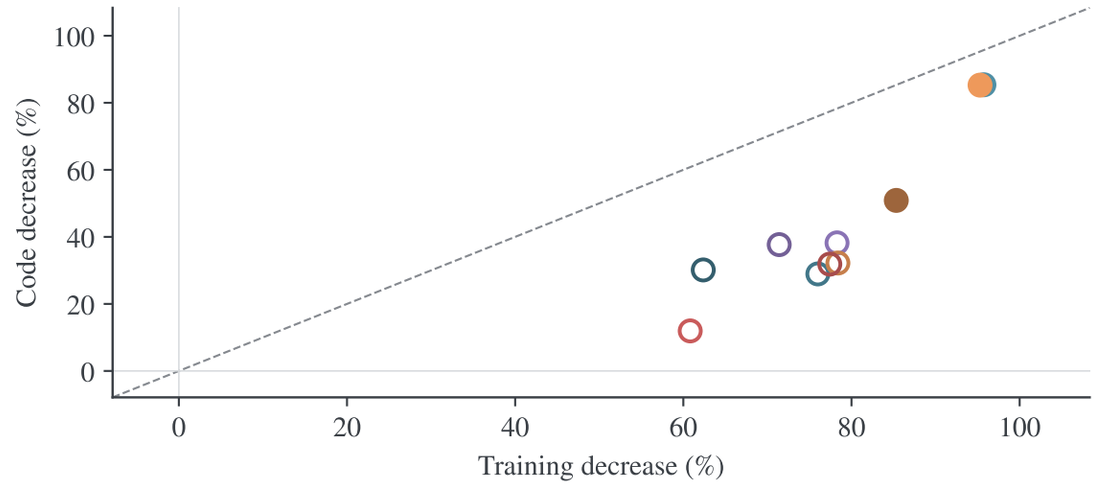
(a) Training vs. Math

n=10; r=0.99; rho=0.90



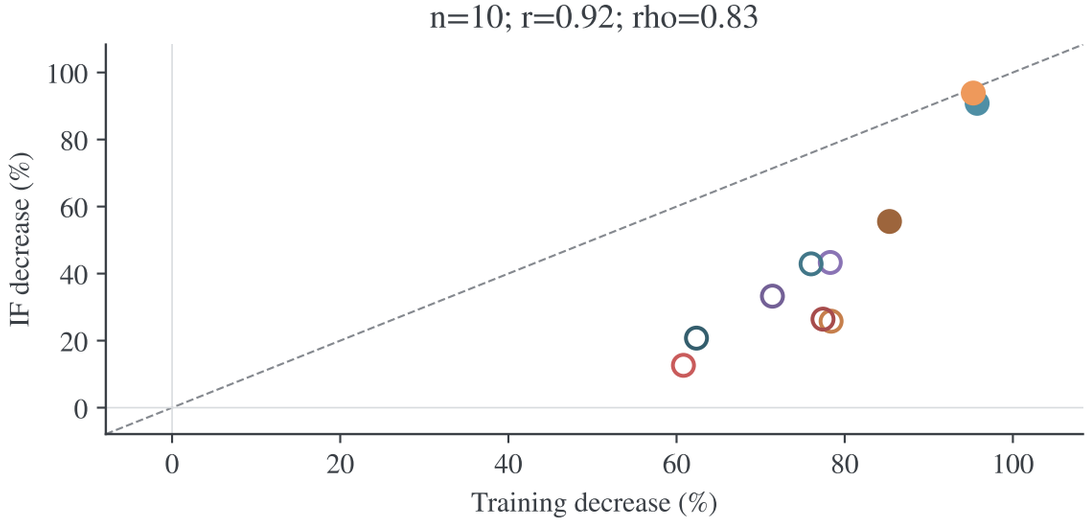
(b) Training vs. Code

n=10; r=0.91; rho=0.88



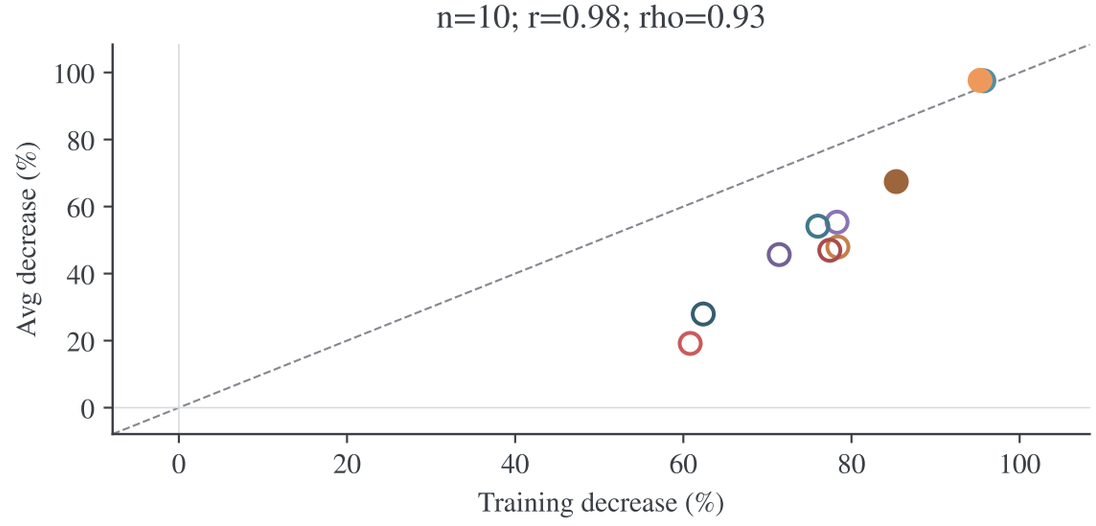
(c) Training vs. IF

n=10; r=0.92; rho=0.83



(d) Training vs. Avg

n=10; r=0.98; rho=0.93



- Co-rewarding-I-Llama-3.2-3B-Instruct-MATH [320]
- GT-Llama-3.2-3B-Instruct-MATH [320]
- Llama-3.1-8B-Instruct / Unsloth [320]
- Llama-3.1-8B-SimpleRL-Zoo [320]
- Llama-3.1-Nemotron-Nano-8B-v1 [320]
- Llama-3.1-Tulu-3-8B [320]
- Llama-3.1-Tulu-3-8B-DPO [320]
- Llama-3.1-Tulu-3-8B-SFT [320]
- Llama-3.2-1B-Instruct / Unsloth [320]
- Llama-3.2-3B-Instruct-arithmetic-full [320]
- Equal decrease

One point = one run endpoint; triangle: 64 updates; circle: 320; filled: same size; open: cross size.

Fixed-H RKL; dashed: equality; correlations pool all displayed endpoints (mixed budgets/sizes).